

PORTLAND CASHMORE, Australia

WMO: 948280

Lat: 38.3147S

Lon: 141.4706E

Elev: 267

StdP: 14.55

Time Zone: 10.00 (AUS)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.8	39.9	32.9	27.6	48.4	34.5	29.5	47.2	31.7	49.9	28.8	50.1	4.4	0	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.2	87.2	64.8	81.6	63.4	76.5	62.4	67.4	79.2	65.7	76.5	64.1	72.8	12.9	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.3	91.0	69.5	62.4	85.1	68.0	60.9	80.7	66.4	32.0	79.3	30.6	76.7	29.4	73.0	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.0	24.1	21.5	DB	33.8	98.7	1.2	4.6	32.9	102.0	32.2	104.6	31.5	107.2	30.6	110.5	(4)
(5)			WB	33.3	72.4	1.2	3.2	32.4	74.6	31.7	76.5	31.0	78.2	30.1	80.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.9	62.5	63.4	60.9	57.1	53.6	50.4	49.6	50.3	52.3	54.4	57.3	59.3	(6)	
	DBStd	6.64	6.16	5.83	5.51	4.56	3.87	3.09	2.95	3.25	3.98	5.07	5.56	5.74	(7)	
	HDD50	141	0	0	0	1	9	31	44	34	14	7	1	0	(8)	
	HDD65	3502	126	93	157	244	354	437	477	454	380	333	246	201	(9)	
	CDD50	2286	387	374	338	213	121	44	31	44	84	144	219	287	(10)	
	CDD65	176	49	47	30	6	1	0	0	0	1	5	14	23	(11)	
	CDH74	1645	467	383	265	58	3	0	0	0	3	67	157	242	(12)	
	CDH80	644	209	157	101	15	0	0	0	0	0	16	45	101	(13)	
(14) Wind	WSAvg	10.6	10.3	10.1	9.8	9.0	9.9	9.8	11.4	11.5	11.8	11.5	10.9	10.9	(14)	
(15) Precipitation	PrecAvg	31.9	1.3	1.0	1.9	2.2	2.9	4.4	4.1	4.0	3.5	2.5	2.0	1.9	(15)	
	PrecMax	39.3	3.8	3.3	6.9	4.5	6.5	8.0	6.5	7.2	5.3	4.2	4.0	4.5	(16)	
	PrecMin	23.6	0.2	0.1	0.2	0.4	0.9	2.3	2.2	1.0	1.5	0.8	0.6	0.6	(17)	
	PrecStd	4.1	0.9	0.8	1.4	1.2	1.6	1.2	1.3	1.6	1.1	1.1	1.0	1.1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.4	93.4	89.8	81.1	70.3	62.1	60.4	66.3	73.0	81.9	85.7	90.5	(19)	
		MCWB	66.5	67.6	63.8	61.0	57.5	54.4	52.6	54.4	58.7	60.5	62.9	64.7	(20)	
	2%	DB	86.0	84.6	81.8	73.8	65.4	59.0	57.6	61.7	67.4	73.8	78.3	80.2	(21)	
		MCWB	64.7	65.4	62.7	58.9	56.0	53.5	51.8	52.4	55.7	58.9	63.3	62.8	(22)	
	5%	DB	78.1	77.8	74.5	68.5	62.1	57.3	56.1	58.6	62.9	68.5	72.0	73.2	(23)	
		MCWB	64.0	64.0	61.8	58.2	55.5	52.7	51.0	51.9	54.5	57.4	61.6	61.9	(24)	
	10%	DB	71.7	71.9	69.0	64.3	59.8	55.7	54.7	56.2	59.5	63.3	66.4	67.8	(25)	
		MCWB	62.8	63.1	60.6	57.3	54.9	51.7	50.1	50.9	52.9	55.7	59.0	60.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.6	70.8	67.3	63.6	59.9	56.7	55.5	56.6	60.9	63.4	67.7	67.5	(27)	
		MCDB	81.8	81.1	79.9	73.3	64.2	59.1	58.0	62.0	67.2	74.7	77.7	80.0	(28)	
	2%	WB	67.0	67.4	64.8	61.4	58.2	54.9	53.2	54.3	57.6	60.4	64.5	65.0	(29)	
		MCDB	78.8	79.6	75.0	68.1	62.2	57.6	55.9	58.9	64.8	70.5	74.0	75.8	(30)	
	5%	WB	64.9	65.4	63.1	59.8	56.8	53.5	52.0	52.9	55.4	58.3	61.9	62.7	(31)	
		MCDB	75.2	74.8	70.7	65.2	60.5	56.2	54.7	57.0	61.2	66.4	70.7	72.5	(32)	
	10%	WB	63.1	63.7	61.4	58.3	55.5	52.3	50.9	51.7	53.8	56.2	59.5	60.7	(33)	
		MCDB	71.5	70.6	67.8	63.3	59.1	55.1	53.8	55.6	58.6	62.2	65.7	67.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	15.5	14.2	14.3	13.7	11.4	10.3	9.9	11.0	12.5	14.1	14.0	14.7	(35)	
		MCDBR	28.3	26.4	25.8	22.3	15.4	12.8	11.6	15.6	19.8	25.4	26.2	26.4	(36)	
		MCWBR	10.6	9.8	10.2	11.0	9.4	9.0	7.9	9.2	11.2	13.0	12.7	11.6	(37)	
	5% WB	MCDBR	23.0	20.6	19.7	17.1	12.3	10.6	10.1	13.0	16.4	21.8	22.7	22.3	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.340	0.339	0.326	0.318	0.306	0.304	0.303	0.314	0.338	0.338	0.336	0.337	(40)	
		taud	2.548	2.559	2.597	2.601	2.614	2.601	2.607	2.571	2.486	2.495	2.529	2.545	(41)	
		Ebn at Noon	315	308	299	282	266	256	265	279	290	304	314	317	(42)	
	Edh at Noon	Edh at Noon	34	32	29	25	21	20	21	25	31	34	35	35	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2250	1915	1504	1053	707	561	615	888	1284	1714	2008	2229	(44)	
	RadStd	RadStd	118	106	76	70	38	36	34	46	87	119	125	145	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.76	N/A	N/A	N/A	+0.81	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	-164	+230	+68

Nomenclature: See separate page